9.3 Microbiology of Water

Aquatic microbiology is the study of microorganisms and microbial communities in water environments. Aquatic environments occupy more than 70% of the earth's surface. Water is essential for life and one of the most important natural resource. Thus, protection and preservation of aquatic environments are vital for the continuation of life. There are two kinds of water found on earth:

1. Salt water (97%)
2. Fresh water (3%)

9.3.1 Salt Water Microflora

Salt water contains a significant level of dissolved salts. The major bodies of salt water are oceans, seas, estuaries, and certain salt water lakes. Average salinity of ocean is 3.5 grams per 100 grams of water. The pH of salt water remains relatively constant at a range of 8.3 to 8.5. The temperature of seawater fluctuates depending on location, season and depth.

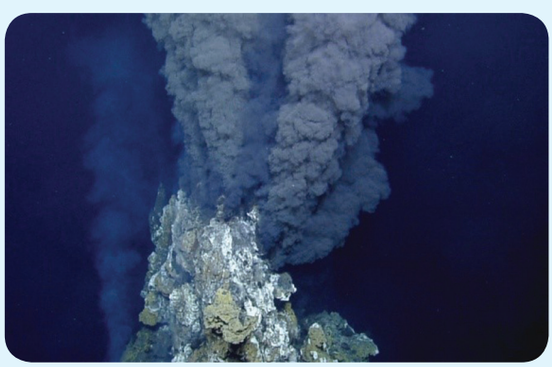
9.3.2 Estuaries

A partly enclosed coastal body of water in which fresh water is mixed with seawater is called an estuary. As a consequence of this geographical location, estuaries are of salinity levels that range from less than 0.5 to 2.5 grams of dissolved salts per 100 grams of water at their mouths, where the water flows into the marine environment. Estuaries are ecologically sensitive environments and serve as habitats for fish and wide variety of marine life. There is an increasing concern about the quality of life in estuaries seriously damaged by human impact, including over development and pollution from industrial and waste water discharges. The bacterial population in estuaries consists of *Pseudomonas, Flavobacterium,* and *Vibrio*, as well as enteric organisms. The quantities and types of organisms vary, and depend on the tide, rainfall, salinity, depth and temperature. Most of the bacteria found in water runoff come from animal and



The hot, sulphur-rich, acidic pool of yellow stone national park (U.S.) is home to many Archaebacteria. The colour differences in the pool result from the different communities of microbes that are able to thrive at extreme water temperatures. *Pyrolobus fumarii* is a unique Archaebacteria, which is hyperthermophilic that can grow at the temperature of 113°C. Some Archaebacteria live in thousands of miles deep in ocean near superheated volcanic vents.





TN_GOVT_XI_Micro_Biology_CH09.indd 128 16-02-2019 10.02.23 AM